Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

1 Tomato

1.1 Vegetative Growth

- **Vegetative growth stages:** Documented vegetative development stages for tomato (*Solanum lycopersicum*) include a 10–day seedling establishment stage after transplanting, followed by a 30–day vegetative growth stage, and a 30–day flower initiation to flowering transition stage. Specific processes regarding root development, stem elongation, branching, leaf development, and canopy formation are Not specified in the provided sources.

- **Environmental requirements:** High maximum temperatures exceeding (40°C)*orreaching* (42°C)*causeincreasedflowerandfruitdrop, necessitatingsufficientirrigationtomitigatethermalstress* $-20mg/Lpromoteshigheryieldunderlowtemperatureconditions.Relativehumidity, sunlight, pho$

- **Soil and growing conditions:** Soil preparation requires incorporation of 15–20 tonnes/ha of well-decomposed farmyard manure into the soil to ensure adequate fertility. Testing of soil pH and electrical conductivity is documented, but specific target ranges, soil texture, drainage, and structural parameters are Not specified in the provided sources.

- **Root development:** Root establishment is supported by providing light irrigation 3–4 days after transplanting and supplying balanced nutrients during the 10–day post-transplant establishment phase. Root growth patterns, root distribution, rooting depth, root health metrics, and nutrient and water uptake mechanisms are Not specified in the provided sources.

- **Stem and leaf development:** Plant height, stem thickness, branching patterns, leaf number, leaf area, canopy development metrics, and visual indicators of healthy vegetative growth are Not specified in the provided sources.

- **Irrigation and moisture management:** Tomato plants are highly sensitive to water application, and light irrigation must be supplied 3–4 days after transplanting. Irrigation intervals should be adjusted according to soil type and rainfall, scheduled at 7–8 days during kharif, 10–12 days during rabi, and 5–6 days during summer. Heavy irrigation following a long period of drought must be avoided to prevent fruit cracking. Under fertigation management for hybrids, daily drip irrigation is supplied for 1 hour, accompanied by water-soluble fertilizer applications split once every 3 days.

- **Nutrient management:** Standard non-hybrid varieties require a recommended dose of 120 kg/ha N, 80 kg/ha

(P_2O_5
), $and 50kg/ha$
(K_2O
), $withhalfofNandfullPandKappliedatplantingasabasaldoseandtheremainginghalfofNtop-$
$dressed30daysaftertransplanting.Hybridvarietiesunderconventionalfertilizationrequire180kg/haN$
(P_2O_5
), $and 60kg/ha$
(K_2O
), $administeredas60kg/haNwithhalfofPandKattransplanting, 60kg/haNwiththeremainingPandKt$
$dressed30daysaftertransplanting, andathirdsplitof60kg/haNtop-dressed50daysaftertransplanting$
$250 : 250kg/haofN, P, andK, where$
(75%
)$ofP($
(187.5 kg/ha
)$P, equivalentto1172kg/hasinglesuperphosphate)isappliedbasallyandtheremaining200 :$
$62.5 : 250kg/haNPKisfertigatedevery3days.Thefertigationregimesupplies$
(10%
)$N($
(20 kg/ha
)),
(5%
)$P($
(12.5 kg/ha
)), and
(10%
)$K($
(25 kg/ha
))$duringthe10 - -dayestablishmentstageusing19 : 19 : 19, 13 : 0 : 45, andUrea($
(46%
)$N), followedby$
(40%
)$N($
(80 kg/ha
)),
(10%
)$P($
(15 kg/ha
)), and
(40%
)$K($
(40 kg/ha
))$duringthe30--dayvegetativestageusing12 : 61 : 0, 13 : 0 : 45, andUrea.SeedtreatmentwithGibberell$
$-100mg/Lenhancesseedgermination.Specificdeficiencysymptoms, toxicitythresholds, andmicronut$

- **Weed management:** Organic mulching in vegetable crops is documented to minimize soil evaporation and maintain soil moisture. Specific weed species, competition thresholds, chemical herbicide recommendations, application timings, and precautions are Not specified in the provided sources.

- **Intercultural operations and plant support:** Mulching is documented to maintain soil moisture and suppress surface evaporation. Specific procedures, timings, and instructions for staking, tying, pruning, sucker removal, training, and soil loosening are Not specified in the provided sources.
- **Pest and disease management:** General crop protection recommendations emphasize integrated pest management, utilization of bio-pesticides, conservation of natural enemies, and withholding chemical sprays at least 8 days prior to fruit harvest. Specific insect pests, mites, nematodes, fungal or bacterial pathogens, causal organisms, diagnostic symptoms, and pesticide prescriptions during vegetative growth are Not specified in the provided sources.
- **Physiological disorders and growth stress:** Heavy irrigation following a prolonged dry period causes fruit cracking. High ambient temperature stress (($T_{max} > 40°C$))$leads to flower and fruit drop, which is managed through adequate irrigation scheduling. Nutrient disord$
- **Growth monitoring and assessment:** Quantitative monitoring parameters including plant height, stem diameter, leaf count, canopy architecture, root system condition, growth rates, uniformity thresholds, and assessment frequencies are Not specified in the provided sources.
- **Growth duration and scheduling:** The transplanting-to-establishment stage lasts 10 days, followed by a 30–day vegetative growth stage and a 30–day flower initiation stage. Conventional top-dressing schedules require field operations at 30 days post-transplanting for standard varieties and at 30 and 50 days post-transplanting for hybrids. Irrigation is scheduled at intervals of 5–6 days in summer, 7–8 days in kharif, and 10–12 days in rabi.
- **Effects on crop performance:** Incorporation of 15–20 tonnes/ha farmyard manure combined with recommended NPK fertilization ensures optimal plant vigor and crop yield. Consistent light irrigation prevents fruit cracking, while supplying sufficient irrigation during high-temperature spells reduces stress-induced flower and fruit loss. Foliar application of Gibberellic acid at 10–20 mg/L enhances yield under low temperature conditions.
- **Integrated management workflow:** Field operations begin with soil preparation and basal incorporation of 15–20 tonnes/ha farmyard manure along with basal inorganic fertilizers. Seedlings are transplanted and given light irrigation after 3–4 days to facilitate establishment during the initial 10–day period, supported by fertigation supplying (10%) N, (5%) P, and (10%) K. The subsequent 30–day vegetative growth phase is managed with fertigation supplying (40%) N, (10%) P, and

(40%
) K or top-dressing 60 kg/ha N at 30 days post-transplanting, combined with surface mulching to conserve moisture. Irrigation is maintained at 5–6 day intervals during summer (and increased during extreme heat above
(40°C
)), $7--8 days during kharif, and 10--12 days during rabi. A second top-dressing of 60 kg/ha N is applied at$
$transplanting for hybrids as plants complete vegetative growth and transition toward flowering.$